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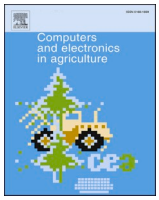

Original papers

Video-based fruit detection and tracking: effects of scanning conditions on fruit load estimation

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ABSTRACT

Automated fruit counting and yield estimation systems are necessary for efficient orchard management. This study presents a computer vision system based on video multi-object-tracking for fruit load estimation in apple orchards and provides a comprehensive analysis of the system performance under diverse scanning conditions. The system integrates fruit detection, tracking, localization within orchard, and fruit load map generation. Experiments were carried out in an experimental apple orchard containing 420 apple trees. Data was collected with two different RGB-D sensors (Azure Kinect DK and ZED 2) at three different scanning distances (125 cm, 175 cm, and 225 cm) on two different dates prior to the harvest. Comparing the two evaluated sensors, Azure Kinect provided more consistent performance across different dates. Results also show that the longer scanning distance improves accuracy due to seeing the full tree view gives better fruit counts than close partial views. Between the two dates, best results were achieved near harvest due to fruit color at this stage, achieving a Mean Absolute Percentage Error (MAPE) of 6.91 % and a determination coefficient (R^2) of 0.733 (using ZED2 sensor at 225 cm distance). Finally, a test comparing scanning from one or both sides of the tree row showed that bilateral scanning improved fruit load estimation at the stretch level by incorporating information from both sides of the canopy. The results of this work demonstrate the effectiveness of the video fruit tracking systems as a useful tool for automating fruit load estimation.

1. Introduction

Automated fruit counting systems are increasingly recognized as valuable tools for improving orchard management. By enabling more accurate and spatially detailed assessment, these systems facilitate better planning of harvest, labour, and storage, contributing to more efficient orchard management (He et al., 2022). However, despite the significant progress in automated solutions, driven by deep learning in computer vision, accurately counting fruits using automated systems remains a challenging task. One of the primary challenges is fruit occlusion (Gené-Mola et al., 2023b; Tian et al., 2019), where fruits are partially or fully hidden by leaves, branches, or other fruits. This issue directly impacts the reliability of computer vision-based fruit counting methods, often leading to underestimation, missed detections, or inconsistent observations when fruits become intermittently visible. Consequently, occlusions affect not only detection recall but also the reliability of automated counting systems. Furthermore, occlusions complicate the development of systems capable of generalizing across diverse orchard environments, limiting their practical deployment (Tang et al., 2023).

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One way to address this challenge is to capture data from multiple viewpoints, assuming that fruits hidden from one angle may be visible from another. However, this method risks overestimating fruit counts due to duplicate detections from overlapping perspectives. To address this issue, some studies have explored 3D fruit detection and localization techniques to eliminate duplicate counts (Gené-Mola et al., 2021, 2020; Santos et al., 2024). Nevertheless, these methods often require sophisticated photogrammetry and precise camera calibration, making them complex to implement. Additionally, 3D based methods use to have a higher computational complexity that limits its use for real time applications, and consequently, prevents its deployment for fruit counting in large commercial orchards (Miranda et al., 2023).

As an alternative, recent studies have proposed video-based fruit detection and counting pipelines based on continuous motion videos along the tree rows. In these systems, fruits are detected in each video frame, and object tracking algorithms are used to prevent double counting, by assigning unique identities to fruits across frames (Ariza-Sentís et al., 2024). Most of these pipelines combine object detection architectures with tracking algorithms and can be categorized into approaches based on single-object tracking (SOT) and those based on multiple-object tracking (MOT). Among the former, the methodology proposed by Gao et al. (2022) combines the YOLOv4-tiny network with a trunk tracking approach, offering advantages in terms of low computational cost. In a related work (Wu et al., 2023b), proposed a method to reduce abnormal fruit detections using distance and confidence thresholds. The same authors also introduced a twice-matched fruit counting pipeline to address the limitations posed by clustered fruits (Wu et al., 2023a).

Regarding MOT-based methods, matching detections of the same object between consecutive frames typically relies on motion and appearance information. SORT (Bewley et al., 2016a) and ByteTrack (Zhang et al., 2022a) use a Kalman filter to predict the object's future position, while DeepSORT (Wojke et al., 2017) incorporates appearance descriptors to associate detections based on both motion predictions and visual similarity. In recent years, MOT tracking algorithms have been used for apple counting, with several compared in a comprehensive study of tracking-by-detection methods by Villacrés et al. (2023). For example, He et al. (2022) combined the apple detection results from YOLO with a cascade matching approach (CascadeSORT), while more recently, Cao et al. (2025) employed drones to detect and count apples under different occlusion and illumination conditions, combining YOLO with DeepSORT. Vision Transformers have also been applied to automatic apple fruit detection by Hu et. (2023), who combined them with YOLO and SORT/CascadeSORT tracking methods. Similar MOT-based counting methods have also been developed for other fruits such as mangoes (Gu et al., 2024), tomatoes (Kang et al., 2024), oranges (Zhang et al., 2022), and strawberries (Zhou et al., 2024).

Many MOT-based fruit counting studies focus mainly on evaluating tracking performance using standard metrics, such as Multiple Object Tracking Accuracy (MOTA) or Higher Order Tracking Accuracy (HOTA). Recent studies have continued to advance MOT-based fruit counting by comparing modern detector-tracker combinations for tomato cluster yield estimation (Qi et al., 2025) and by improving Kalman-filter-based multi-object tracking for fruit counting in complex orchard scenes (Zhai et al., 2025). However, discrepancies between predicted fruit counts and actual harvest data are still less frequently addressed. This gap underscores the need to evaluate video-based fruit counting systems not only through detection or tracking metrics, but also through their ability to estimate actual fruit load under practical orchard acquisition conditions. The present study extends a previous work (Gené-Mola et al., 2023a) which evaluated the performance of MOT algorithms such as SORT, DeepSORT, and ByteTrack for fruit tracking in orchards. Building upon that tracker evaluation, the contribution of this work lies in the development of a fruit counting algorithm based on the tracking output and a comprehensive analysis of the counting system's performance under different scanning conditions.

This study hypothesizes that although closer scanning distances improve fruit detection, they may compromise overall counting accuracy due to limited tree coverage within the camera field of view. Therefore, achieving a balance between scanning closer for better detection and scanning farther for broader tree visibility is essential for accurate fruit counting.

To address this, the current work investigates how scanning conditions impact MOT-based fruit counting accuracy, considering factors such as sensor type, fruit maturity stage (green versus mature), and scanning distance. The proposed end-to-end system is evaluated using data acquired on different dates, with different RGB-D sensors and at multiple sensor-to-row distances, enabling a systematic assessment of robustness beyond a single controlled setup. By explicitly analysing how acquisition conditions influence the trade-off between detection reliability and canopy visibility, this work provides insight into practical deployment constraints in orchard environments.

Furthermore, a key contribution of this study is the direct comparison between estimated fruit counts and actual harvest data, enabling the creation of load maps. This harvest-level ground truth provides a reliable basis for quantitative evaluation and offers a practical perspective, moving beyond standard video detection metrics. Another strength of this work is the comprehensive evaluation of the proposed system under multiple field acquisition conditions, including different RGB-D sensors, acquisition dates, scanning distances, and scanning patterns. Together, these aspects provide actionable insights for improving fruit counting methodologies in real-world orchard environments.

The rest of this paper is organized as follows: Section II describes the experimental setup used to create the dataset, outlines the methodology pipeline, and explains the evaluation metrics. Section III presents the results of fruit counting under various scanning conditions. Section IV discusses the findings, highlighting the scanning conditions most favourable for fruit counting and comparing the results with other state-of-the-art approaches. Finally, Section V concludes the study by summarizing key insights and proposing directions for future research.

2. Materials and methods

2.1. Experimental setup

The data were acquired from an experimental mature apple orchard (Story® Inored^{cov}), located in the municipality of Mollerussa, Catalonia, Spain (latitude: 41°37′04.2″N; longitude: 0°52′13.4″E) and owned by the Institute of AgriFood, Research and Technology (IRTA). In the studied orchard, the trees were trained using a fruiting wall system, planted at a density of 3.6 m × 1 m, with a maximum canopy height of 3.5 m.

Fig. 1a shows a top view of the study area, which consists of four rows of apple trees, each 105 m long and containing a total of 420 trees. The rows were segmented into 21 stretches of 5 m each, giving a total of 84 stretches. Fig. 1b presents a top view of the orchard, where the number of apples harvested (October 5th, 2021) in each 5-metre section is represented by a colour gradient. The ground truth for the fruit load map was obtained by individually harvesting each section and counting the number of fruits per stretch with a commercial apple sorting line machine (Maf Roda Agrobotic, Montauban, France). Stretches 4, 5 and 6 of row 2 and stretch 5 of row 4 are not coloured because the ground truth of these sections is not available.

The acquisition equipment included two RGB-D cameras, a ZED 2 stereo camera (Stereolabs, San Francisco, CA, USA) and an Azure Kinect DK time-of-flight (ToF) camera (Microsoft Corporation, Redmond, WA, USA), both at a height of 162 cm from the ground, as well as an Ardusimple simpleRTK2B (Ardusimple Co., Ltd., Lleida, Spain) global navigation satellite system (GNSS) placed at a height of 207 cm from the ground. Table 1 presents the main characteristics of the RGB-D cameras. All the sensors were controlled with a Windows operating system laptop, running the AK_ACQS software (Miranda et al., 2022) for the data acquisition. This equipment was mounted on a vertical mast boarded on

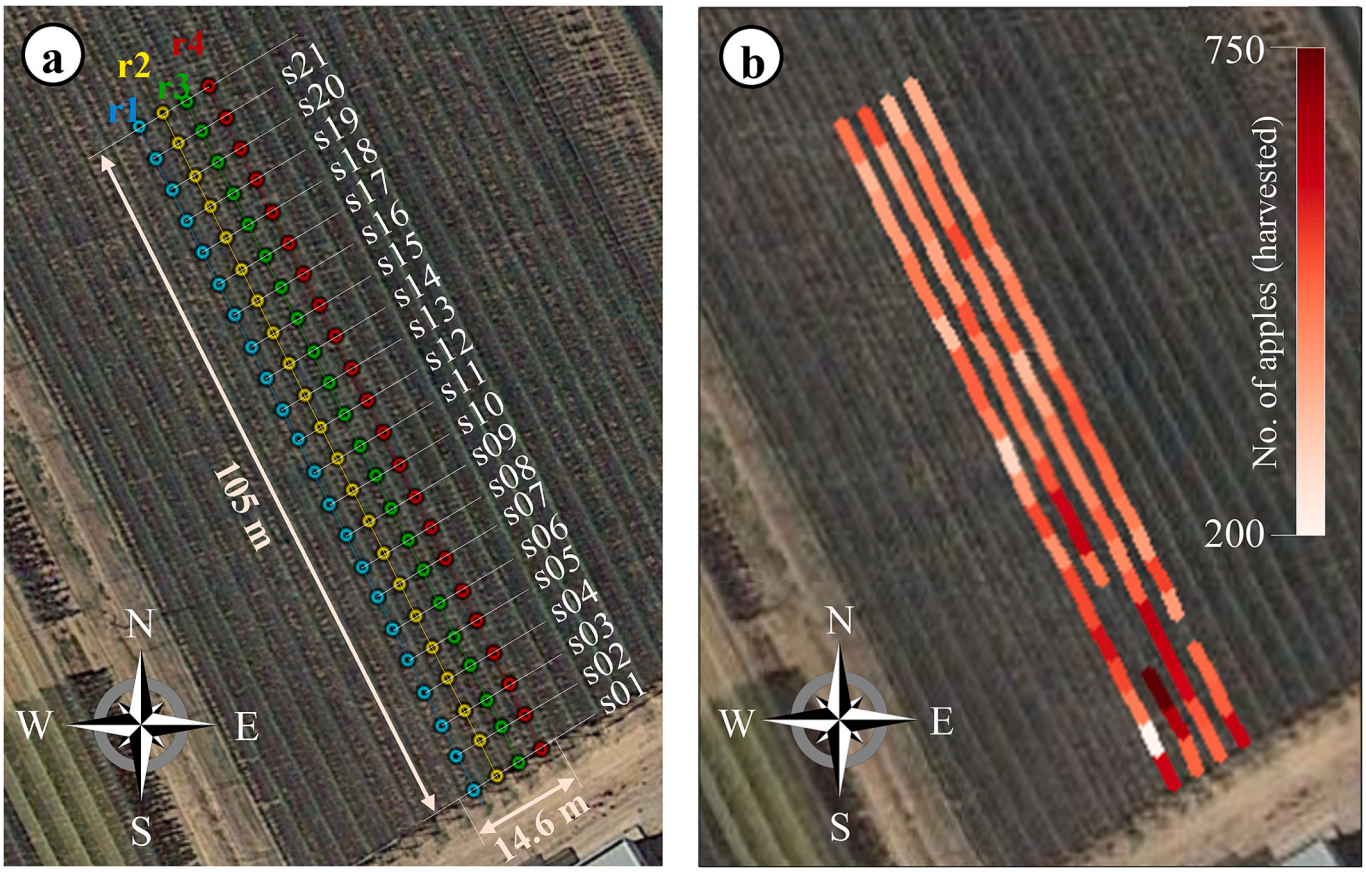

Fig. 1. a) Top view of the orchard showing scanned rows and stretches used to evaluate fruit counting (adapted from Gené-Mola et al., 2023a). rX indicates the number of row. sX indicates the number of stretch. b) Ground truth map of the fruits counted in each stretch.

Table 1
Specifications of the ZED 2 and Azure Kinect RGB-D cameras.

	ZED 2	Azure Kinect
Principle	Stereo vision	Time of Flight (ToF)
RGB resolution (px)	1920 × 1080	1920 × 1080
RGB field of view (°)	110 × 70	90 × 59
Depth resolution (px)	1920 × 1080	640 × 576
Depth field of view (°)	110 × 70	75 × 65
Depth Range (m)	0.3–20	0.5–3.86
Operating frequency (fps)	30	30

an electric all-terrain vehicle as shown in Fig. 2.

Two sets of data were collected on different dates. The first data collection was carried out four weeks before harvest, on September 6th, 2021, when apples featured a two-tone hue of pink and green (Fig. 3a), and the second data collection was carried out one week before harvest on September 28th, 2021, when apples had reached a fully red coloration (Fig. 3b).

Data acquisitions were performed at three different scanning distances, placing the sensors at 1.25 m, 1.75 m, and 2.25 m from the longitudinal axis of the tree row. Shorter distances allow for better visibility and detail of the apples but limit the field of view, making it difficult to capture the entire canopy structure. In contrast, greater distances provide a more complete view of the tree but reduce the resolution and visibility of individual fruits. Therefore, an analysis of the balance between fruit detail and overall tree coverage is required. Each row was scanned twice, from the east and from the west side. To capture the data, the scanning system was driven at a constant speed of 2 km/h along the orchard alleyways.

2.2. Methodology pipeline

Fig. 4 presents a flow chart that outlines the complete workflow from data acquisition to yield map generation. The process starts with video acquisition using two RGB-D cameras: Azure Kinect DK and ZED 2. For each video frame, the system performs fruit detection and tracking, assigning a unique identifier (ID) to each apple across successive frames. Once all the frames have been processed, the output is a list of detected apples with their associated tracking IDs and spatial information. For validation purposes, this information is then processed by the stretch assignation module, which associates each apple with a specific stretch of the orchard row based on its location. This step enables comparison with the stretch-level harvested ground truth. With all the apples already assigned to their corresponding stretches, the system performs a fruit count per stretch. Finally, an orchard block-level yield map is generated, representing the fruit count of each 5-m stretch arranged according to its row and position along the row.

2.3. Detection and tracking

The YoloV5x (Ultralytics, 2022) convolutional neural network (CNN) was used in this work to detect fruits in each frame of the RGB video sequences captured with the Azure Kinect DK and ZED 2.

The YoloV5x architecture was initially trained on the COCO (Lin et al., 2014) dataset and then fine-tuned using our custom dataset. For the fine-tuning process, we used Stochastic Gradient Descent (SGD) with a learning rate of 0.01, a weight decay of 0.0001, and a batch size of 4. All training and inference processes were conducted on a workstation equipped with an Intel Core i7-9800X CPU (3.8 GHz), 32 GB of RAM, and two NVIDIA GeForce RTX 2080 Ti GPUs (11 GB VRAM each), running Ubuntu 20.04.6 LTS (64-bit) with CUDA 11.1.

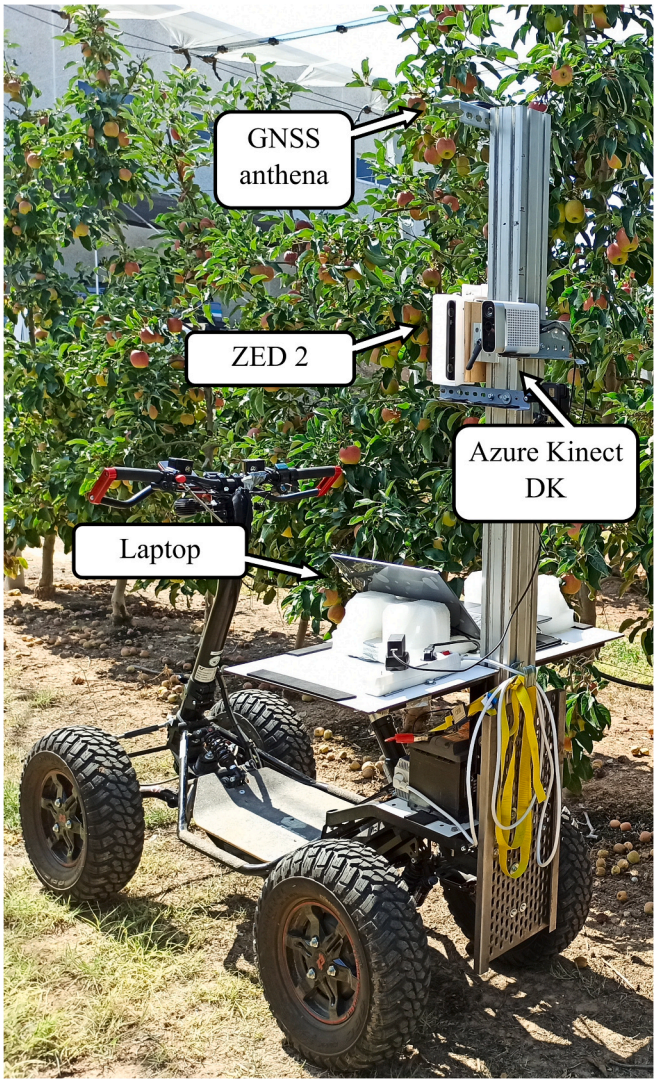

Fig. 2. Orchard scanning equipment.

To create our custom dataset, a total of 1723 frames from stretches 4, 5 and 6 of row 2 were annotated, with 1295 frames used for training, 203 for validation, and 225 for testing. This dataset includes video-sequences at all evaluated conditions (different sensors, distances and dates). Using the Supervisely platform (https://supervisely.com/), unique apple IDs were manually assigned to the bounding boxes of all visible apples in these frames. The accuracy of apple counting was evaluated by comparing the number of fruits detected across rows 1 to 4 with the total number of harvested fruits. To avoid data leakage between training and evaluation, stretches 4, 5 and 6 of row 2 were excluded from the fruit counting experiments. Consequently, the counting performance was evaluated on different rows and stretches of the orchard, allowing the performance of the detection model to be assessed on unseen spatial regions within the same orchard.

The fruit tracking task involved a comparison of three different trackers: SORT (Bewley et al., 2016b), DeepSORT (Wojke et al., 2017) and ByteTrack (Zhang et al., 2022b). These trackers follow the tracking-by-detection paradigm, in which object detections obtained in each frame are associated over time to generate object trajectories. SORT is a motion-based tracker that predicts the position of existing tracks using a Kalman filter and associates them with new detections using the Hungarian algorithm. This approach is computationally efficient, but it relies only on object motion and spatial overlap, without considering visual appearance. DeepSORT builds upon SORT by integrating object appearance information extracted from a re-identification (Re-ID) network (Wojke and Bewley, 2018), which is used together with motion information to improve the association between detections and previous trajectories. This can reduce identity switches in scenarios where objects have distinctive visual features, although it also increases computational cost. Both algorithms (SORT and DeepSORT) classify the detections into high and low reliability based on the score provided by the object detector (YOLOv5). Highly reliable detections are assigned to active trajectories using a Kalman filter motion model and the Hungarian algorithm applied on distance measurements. Alternatively, ByteTrack introduces an improvement over SORT and DeepSORT by incorporating a second association step. This additional step helps the algorithm associate low-confidence detections with unassigned trajectories, maximizing the algorithm's robustness in scenarios where objects are partially. The performance of these trackers was evaluated on nine video sequences containing a total of 1723 video frames from stetches 5 and 6 of row number 2 and recorded under the different experimental conditions described above.

2.4. Stretch-based assignment for validation and fruit counting

Since the ground truth yield map was generated by individually harvesting and counting fruits within predefined 5-meter stretches (Section 2.1), a stretch-level comparison was required to quantitatively evaluate the accuracy of the proposed counting system. Therefore, for validation purposes, each detected apple in the video sequences was assigned to its corresponding stretch to compare predicted and actual fruit counts at the same spatial resolution.

To visually identify the boundaries between stretches in the video frames, a piece of red and white tape was placed at each stretch transition. These markers, referred to as *change marks*, indicate the precise locations where the transition from one stretch to another occurs. It is to be noted that the use of physical markers and the manual annotation of change frames were implemented exclusively to ensure an accurate alignment between video-based detections and the harvested stretch-level ground truth. This step is not required for the operational deployment of the proposed fruit counting system. In commercial applications, fruit load is typically estimated continuously along the row or aggregated at row/orchard level, without predefined 5-meter segmentation.

Fig. 5 illustrates three representative frames in which a change mark is visible. In each image, the change mark is highlighted with a vertical green line. Fig. 5a shows to the first frame in which the change mark between stretches 1 and 2 of row 2 becomes visible. Fig. 5b shows the frame in which the same change mark is aligned with the centre of the image (considered the reference *change frame*). Finally, Fig. 5c shows the last frame in which the change mark is still visible.

All the images between the first (Fig. 5a) and the last (Fig. 5c) in which the change mark appears are crucial for correctly assigning each apple to its corresponding stretch. These transitional frames may contain apples belonging to two adjacent stretches, so it is essential to determine on which side of the change mark each detected apple is located in order to assign it correctly.

The following algorithms were designed to support the stretch-level validation process and are not intrinsic components of the detection and tracking pipeline. Algorithm 1 presents the pseudocode that explains the stretch assignment method. The main objective is to identify the range of frames in which the change mark is visible in the image and its position. The frames where each change mark was aligned with the centre of the image (e.g. Fig. 5b) were manually identified and annotated. These reference frames, termed *change frames*, were provided as prior information to the algorithm.

Given each stretch change frame (Fig. 5b), the algorithm estimates the position of the change mark across all frames where it is visible (from Fig. 5a to Fig. 5c). This is achieved by estimating the frame-by-frame image motion, expressed in pixels per frame. To do this, the

Fig. 3. Apple tree images acquired at three different scanning distances: a) four weeks before harvest; b) one week before harvest.

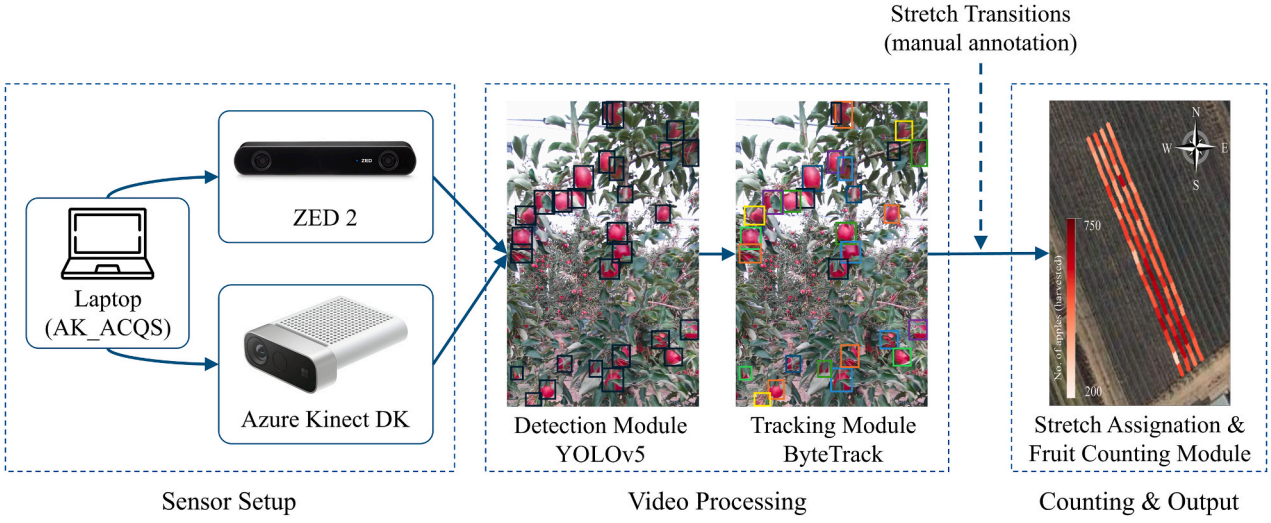

Fig. 4. Flow chart from video capture to orchard block-level yield map.

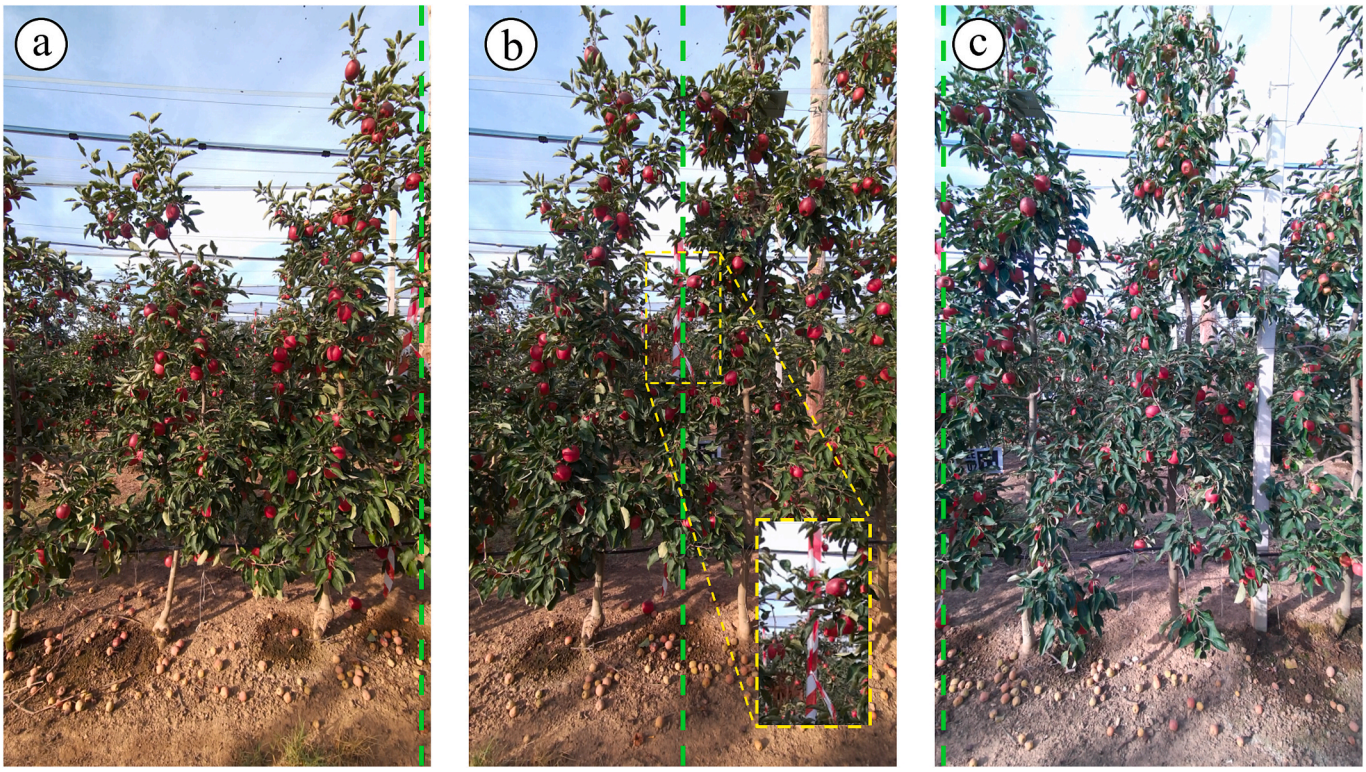

Fig. 5. Set of frames in which the change mark (highlighted with a green dashed vertical line) is seen: a) initial appearance of the change mark; b) frame where the change mark is aligned with the image center (yellow dashed window provides a zoom-in); c) final frame where the change mark is visible.

displacement of all detected apples between consecutive frames is computed, and the average pixel displacement is used as an estimate of the relative frame speed.

This average motion is then used to propagate the position of the change mark forward and backward in the sequence, updating its location in subsequent frames until the change mark is no longer visible in the field of view. This iterative procedure allows for accurate localization of the stretch boundary in all relevant frames, enabling the correct assignment of each detected apple to its corresponding stretch.

Algorithm 1 Compute Change Frames

```
procedure ComputeChangeFrames(apple_data, change_frame_data)
  adjusted_change_frame ← [],
  for each change_frame in change_frame_data do
    apples_in_change ← []
    for each apple in apple_data do
      if apple.framenumber is within change_frame then
        Append apple to apples_in_change
      end if
    end for
    total_velocity, count ← 0, 0
    for each apple in apples_in_change do
      if previous and current frame data available for apple then
        delta_position ← current_position – previous_position
        delta_frame ← current_frame – previous_frame
        velocity ← delta_position/delta_frame
        total_velocity ← total_velocity + velocity
        count ← count + 1
      end if
    end for
    mean_velocity ← if count > 0 then total_velocity/count else 0
    adjusted_start ← change_frame.start
    adjusted_start ← adjusted_start – (mean_velocity × time_backward)
    adjusted_end ← change_frame.end
    adjusted_end ← adjusted_end +(mean_velocity × time_forward)
    adjusted_frame ← (adjusted_start, adjusted_end)
    Append adjusted_frame to adjusted_change_frames
  end for
end procedure
```

Algorithm 2 describes the second step of the stretch assignment method. In the range of frames where a stretch change mark is visible (as determined in Algorithm 1), the position of each apple is analysed relative to the estimated position of the change mark. For each apple tracked in these frames, it is recorded whether it appears on the left or right side.

Since the apple is tracked along different frames, each apple is then assigned to the stretch where it appears most frequently across these frames. This majority-vote strategy addresses the issue of inconsistent apple positions caused by perspective distortion—where an apple near the change mark might appear on different sides of the change mark in consecutive frames.

Finally, if an apple is not visible in any of the frames where a change mark is detected, it is directly assigned to the stretch in which it is observed throughout the video.

Algorithm 2 Assign Apple to Stretch

```
procedure AssignAppleToStretch(apple_data, adjusted_change_frames,
apple_assignments)
  for each apple in apple_data do
    left_count, right_count ← 0, 0
    for each adjusted_change_frame in adjusted_change_frames do
      if apple.last_frame_number is in adjusted_change_frame then
        if apple.position is left of change_frame then
          left_count ← left_count + 1
        else
          right_count ← right_count + 1
        end if
      end if
    end for
    final_stretch ← 'Left' if left_count > right_count else 'Right'
    Append (apple, final_stretch) to apple_assignments
  end for
  return apple_assignments
end procedure
```

Once all detected apples were assigned to their corresponding stretches (as described in Algorithms 1 and 2), a linear regression model was used to correct systematic observation biases in the raw fruit counts. In orchard environments, the number of fruits detected in video sequences does not necessarily correspond to the true number of fruits present on the trees due to factors such as occlusions, fruits visible from both sides of the row, and tracking inconsistencies (e.g., broken tracks or identity switches). Therefore, the regression model relates the number of tracked fruits to the actual number of harvested apples, enabling a more reliable estimation of fruit load at the stretch level. It should be noted that the detections obtained from the east and west scans were processed independently, and no object-level fusion or cross-view matching was performed to explicitly identify the same fruit observed from both sides of the row. Consequently, fruits visible from both viewpoints may be counted twice in the intermediate observations. In the proposed framework, these duplicate observations are not explicitly removed at the instance level but are statistically compensated through the regression model applied.

To ensure the robustness of the estimation and to prevent the linear model from being trained with the same data used for its evaluation, the cross-validation procedure explained in Gené-Mola et al. (2023a) was applied. With this procedure the linear model was evaluated using a leave-one-row-out cross-validation strategy, where the model for each row was trained using data from the remaining rows. Then, performance metrics were calculated for each row and averaged to obtain robust estimations of the model performance. This approach ensures that the resulting metrics are not biased by the specific characteristics of individual rows.

It should be noted that this leave-one-row-out cross-validation assesses the generalization of the regression model across rows within the studied orchard, but not its transferability to other orchards. The fitted coefficients should therefore be regarded as calibration parameters specific to the present orchard and acquisition setup (sensor, scanning distance, date and side). When transferring the system to a new orchard or to different acquisition conditions, re-calibration of the regression model would be required, which in practice could be performed by manually counting the fruits of a reduced number of reference stretches or trees at harvest.

2.5. Evaluation metrics

The evaluation of fruit detection performance is based on several metrics: Precision (P), Recall (R), F1-score (F1) and mean average Precision (mAP). P is the ratio of true positives (correct detections) to the total number of detections made by the model. R represents the percentage of correctly detected fruits compared to the total number of annotated fruits in the ground truth. F1 is the harmonic mean between P and R, it provides a balance between both metrics. mAP is determined by calculating the area under the Precision-Recall (P-R) curve across different Intersection Over Union (IoU) thresholds.

For the evaluation of fruit tracking, several metrics are used to measure the performance of the tracker: Multiple Object Tracking Accuracy (MOTA), Identification F1 (IDF1) and Higher Order Tracking Accuracy (HOTA). Each of these metrics evaluates different aspects of tracking, allowing for a more complete view of the algorithm's effectiveness. MOTA takes into account three types of errors: false positives, false negatives and identity changes, and provides an overall evaluation of the tracker's accuracy based on these cumulative errors. IDF1 is the ratio of correctly identified detections over the average of the total number of objects in the ground truth and the detections made by the model (Ristani et al., 2016). HOTA extends the MOTA concept by integrating multiple evaluation factors. It breaks down the performance of the tracker into different levels, measuring recall and accuracy in detecting and associating identities, and accuracy in locating objects (Luiten et al., 2021).

The performance of the overall fruit counting system is evaluated in terms of mean absolute error (MAE), mean absolute percentage error (MAPE), root mean square error (RMSE) and the coefficient of determination (R^2). The MAE measures the average magnitude of errors between the predicted and actual fruit counts by stretch. MAPE quantifies the error as a percentage of the actual values. The RMSE weighs larger errors more heavily, highlighting the impact of significant deviations between predictions and actual values. The R^2 determines the quality of the model to replicate the results, and the proportion of variation in the results that can be explained by the model.

For clarity and reproducibility, Table 2 summarizes the mathematical formulation of all evaluation metrics used in this study. Where TP, FP and FN denote true positives, false positives and false negatives, respectively; AP is the area under the Precision–Recall curve, computed in practice using a discrete approximation over recall levels; N is the number of classes (N = 1 in this study); IDSW represents identity switches; GT is the total number of ground-truth objects; IDTP, IDFP and IDFN denote true positive, false positive and false negative identity associations; Det.A and Ass. A correspond to detection and association accuracy components of HOTA; y_i and $\hat{y}_i$ are the observed and predicted fruit counts for stretch i, $\bar{y}$ is the mean observed value, and n is the number of evaluated stretches.

3. Results

3.1. Fruit detection and tracking

The YOLOv5x object detection architecture was trained for a total of 273 epochs. The validation loss achieved its minimum value at this stage, indicating that the network had converged optimally. In terms of performance, the network achieved an F1-score of 0.834 and a mAP@0.5 of 0.871, both computed using a Intersection over Union (IoU) threshold of 0.50. These values reflect the high capability of the architecture to correctly detect and classify objects within the evaluated dataset and are in accordance with the cutting-edge results in fruit detection (Koirala et al., 2019; Zhu et al., 2024).

Table 3 shows a detailed comparison of the three evaluated tracking algorithms. These results are obtained from evaluating the three algorithms on validation data corresponding to stretches 5 and 6 of row

Table 2
Mathematical formulation of the evaluation metrics used for detection, tracking and counting performance assessment.

Task	Metric	Formula
Detection	Precision (P)	$\frac{TP}{TP+FP}$
	Recall (R)	$\frac{TP}{TP+FN}$
	F1-score (F1)	$2\times\frac{P\times R}{P+R}$
	Average Precission (AP)	$\int_0^1 P(R)dR$
	mean Average Precision (mAP)	$\frac{1}{N}\sum_{i=1}^{N} AP_i$
Tracking	MOTA	$1-\frac{FN+FP+IDSW}{GT}$
	IDF1	$\frac{2\times IDTP}{2\times IDTP+IDFP+IDFN}$
	HOTA	$\sqrt{Det.A\times Ass.A}$
Counting	MAE	$\frac{1}{n}\sum_{i=1}^{n}\lvert y_i-\hat{y}_i\rvert$
	MAPE	$\frac{100}{n}\sum_{i=1}^{n}\left\lvert\frac{y_i-\hat{y}_i}{y_i}\right\rvert$
	RMSE	$\sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i-\hat{y}_i)^2}$
	R^2	$1-\frac{\sum(y_i-\hat{y}_i)^2}{\sum(y_i-\bar{y})^2}$

Note: TP = true positives; FP = false positives; FN = false negatives; IDSW = identity switches; IDTP = identity true positives; IDFP = identity false positives; IDFN = identity false negatives; GT = ground-truth; Det.A = detection accuracy; Ass.A = association accuracy.

Table 3
Comparison of tracking performance for SORT, DeepSORT, and ByteTrack algorithms in terms of MOTA, IDF1, HOTA, and execution time.

Tracker	MOTA	IDF1	HOTA	Time [ms/img]
SORT	0.6401	0.8091	0.6804	**15.3**
DeepSORT	0.5735	0.7646	0.6824	128.0
ByteTrack	**0.6817**	**0.8369**	**0.6894**	15.4

number 2. For each metric, the highest performance score is highlighted in bold. ByteTrack achieved a MOTA score of 0.6817, demonstrating an advantage over SORT (0.6401) and DeepSORT (0.5735). A similar trend is seen in the IDF1 metric, where ByteTrack performed better than SORT and DeepSORT, indicating that it is better maintaining the identity of the tracked objects across frames. HOTA scores, which balance detection and association accuracy, showed more closely results, ByteTrack achieved a higher score of 0.6894 compared to SORT's 0.6804 and DeepSORT's 0.6824. The accuracy of the object detector has a major impact on the performance of these tracking algorithms. Therefore, the robustness and accuracy of the object detector are critical prerequisites for achieving high quality tracking results.

In terms of computational efficiency, both SORT and ByteTrack showed comparable processing times per image, with averages of 15.3 ms and 15.4 ms, respectively. Conversely, DeepSORT recorded a much higher average processing time of 128.0 ms per image. This increase in the processing time is probably due to the computational overhead associated with the feature extraction and the matching processes in its Re-ID network.

According to these findings, ByteTrack stands out as the most effective tracking methodology among the three evaluated, under the experimental conditions of this study. Its higher MOTA, IDF1, and HOTA values indicate not only better overall tracking accuracy, also improved identity preservation and association quality. This improvement can be attributed to ByteTrack's two-stage association strategy, in which high-confidence detections are first matched to existing tracks and lower-confidence detections are subsequently used to recover unmatched trajectories. In the context of apple fruit tracking, this strategy is particularly relevant because valid fruit detections may receive lower confidence scores due to partial occlusions, changes in viewing angle or illumination variability. By exploiting these lower-confidence detections instead of directly discarding them, ByteTrack can reduce missed detections and fragmented trajectories.

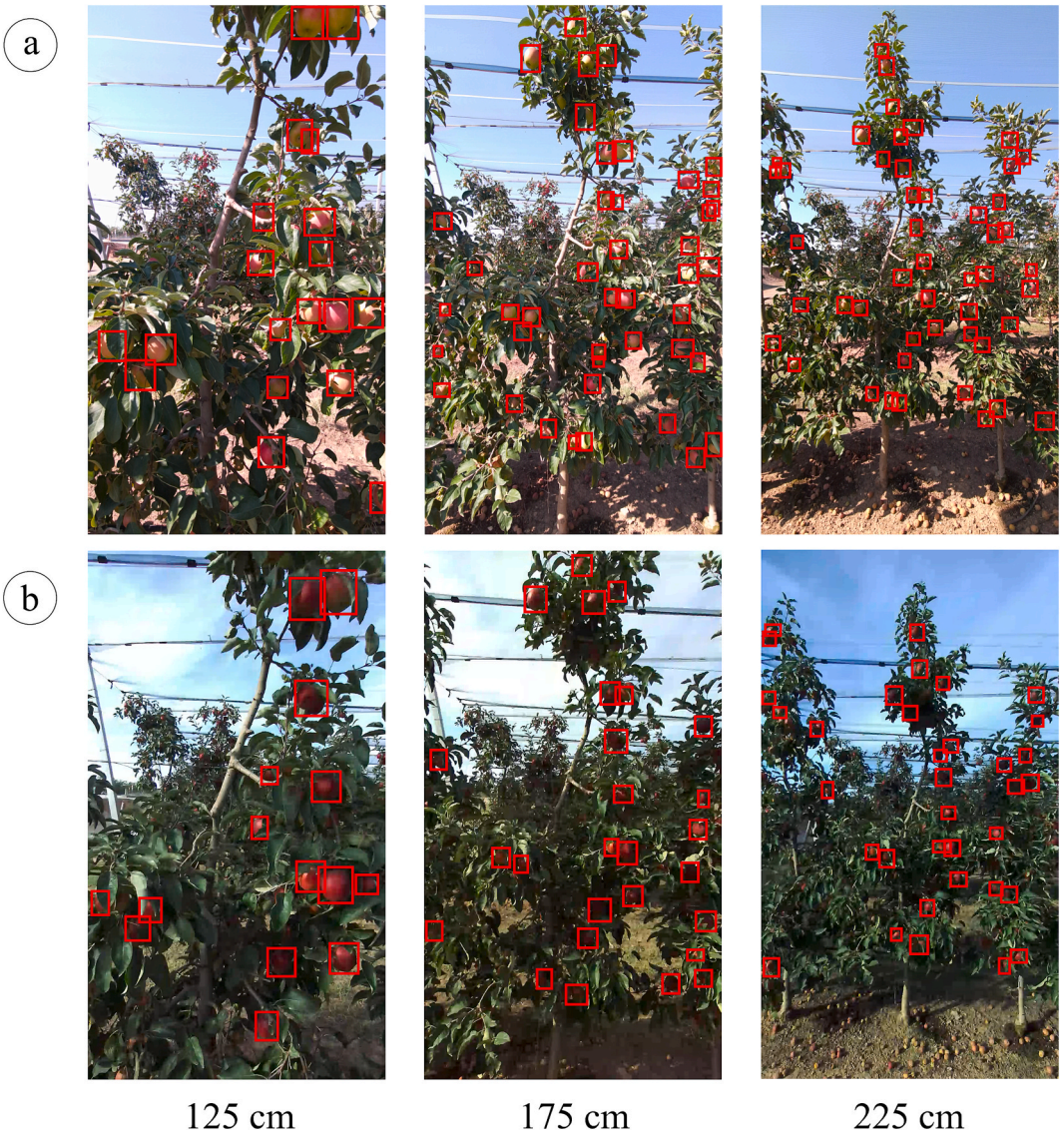

Fig. 6. Apple detection at three different scanning distances with two Sensors on day 28/09: a) Azure Kinect DK; b) ZED 2.

3.2. ZED 2 vs Azure Kinect at different scanning distances and at different dates

Fig. 6 provides a qualitative comparison of apple detection results under different scanning configurations, including variations in RGB-D sensor (ZED 2 vs Azure Kinect) and scanning distance (1.25 m, 1.75 m, and 2.25 m). The figure shows that while apples appear more detailed and with higher resolution at closer distances, the overall detection performance remains relatively consistent across the different distances. This suggests that despite the reduced visual resolution at longer distances, the YOLOv5x-based detection model maintains a robust detection capability. Additionally, the Azure Kinect sensor shows more consistent performance across varying conditions, which supports the quantitative results presented in the next sections.

Fig. 7 shows the relationship between the raw fruit counts obtained from the tracking system and the number of harvested apples for both sensors. Each point corresponds to a five-tree stretch, comparing the number of unique fruit IDs observed in the video sequences with the ground-truth fruit count obtained during harvest.

A clear positive correlation can be observed between the raw counts and the harvested fruit counts for both sensors (Azure Kinect: $R^2 = 0.702$; ZED 2: $R^2 = 0.744$). This indicates that the video-based observations capture the spatial variability of fruit load across the orchard, even before applying the regression-based correction described in Section 2.4.

From this point onward, all reported fruit counting results correspond to the predictions obtained after applying the regression-based bias correction described in Section 2.4.

Table 4 presents the results obtained when evaluating the fruit counting accuracy with the Azure Kinect and the ZED 2 sensors under different experimental conditions, including the date of the data acquisition and the distance between the sensor and the row of trees.

The results for the Azure Kinect sensor show that the highest fruit count accuracy is obtained when the sensor is placed at 225 cm from the row of trees. In this configuration, the MAPE remains below 8 % on both acquisition dates, suggesting a low percentage error relative to the other distances. The RMSE values for this distance are the lowest observed for both dates, indicating a lower dispersion of counting errors. Also, MAE for this distance is the lowest, suggesting a low margin of error relative to the lower distances. In comparison, when reducing the sensor distance to 125 cm, the MAPE increases, exceeding 10 %. A similar behaviour is observed for MAE and RMSE, with errors increasing as we get closer, especially at distances of 125 cm, relative to 225 cm. These results indicate that the accuracy of the Azure Kinect sensor is more sensitive to the distance of acquisition than to the date of data collection. In addition, having images with a lower FoV could be affecting the tracking by generating more tracklets.

Table 4
Fruit counting accuracy with Azure Kinect and ZED 2 sensors at different acquisition dates and sensor-to-tree distances.

Sensor	Date	Distance (cm)	MAE (apples)	MAPE (%)	RMSE (apples)
Azure Kinect	06/09/2021	225	**35.28**	**7.67**	**46.64**
		175	38.33	8.24	52.37
		125	47.37	10.10	62.80
	28/09/2021	225	**35.07**	**7.48**	**46.98**
		175	36.97	7.69	50.40
		125	46.68	10.00	61.90
ZED 2	06/09/2021	225	**42.98**	**9.44**	**57.86**
		175	46.51	10.00	61.85
		125	52.76	11.82	70.21
	28/09/2021	225	**31.71**	**6.91**	**43.04**
		175	36.76	7.71	50.10
		125	43.42	9.29	55.46

For the ZED 2 sensor, the highest fruit count accuracy is also obtained when the sensor is placed at a distance of 225 cm from the row of trees. However, it is observed that in this case, the acquisition date factor has a direct effect on the accuracy of the counting system. In the data obtained on 06/09, for the different distance configurations, the values in the three metrics are higher than those from the Azure Kinect, indicating greater error and a less adequate configuration. Comparing the two acquisition periods for the ZED 2 sensor, the results show higher error when the count is done 4 weeks before harvest. This behaviour is also observed with the Azure Kinect sensor, although to a lesser extent. In the best distance configuration, at 225 cm, all three metrics decrease, achieving an improvement in MAPE of 2.53 %, indicating a lower dispersion in the count.

Fig. 8 shows graphically the value of the determination coefficient

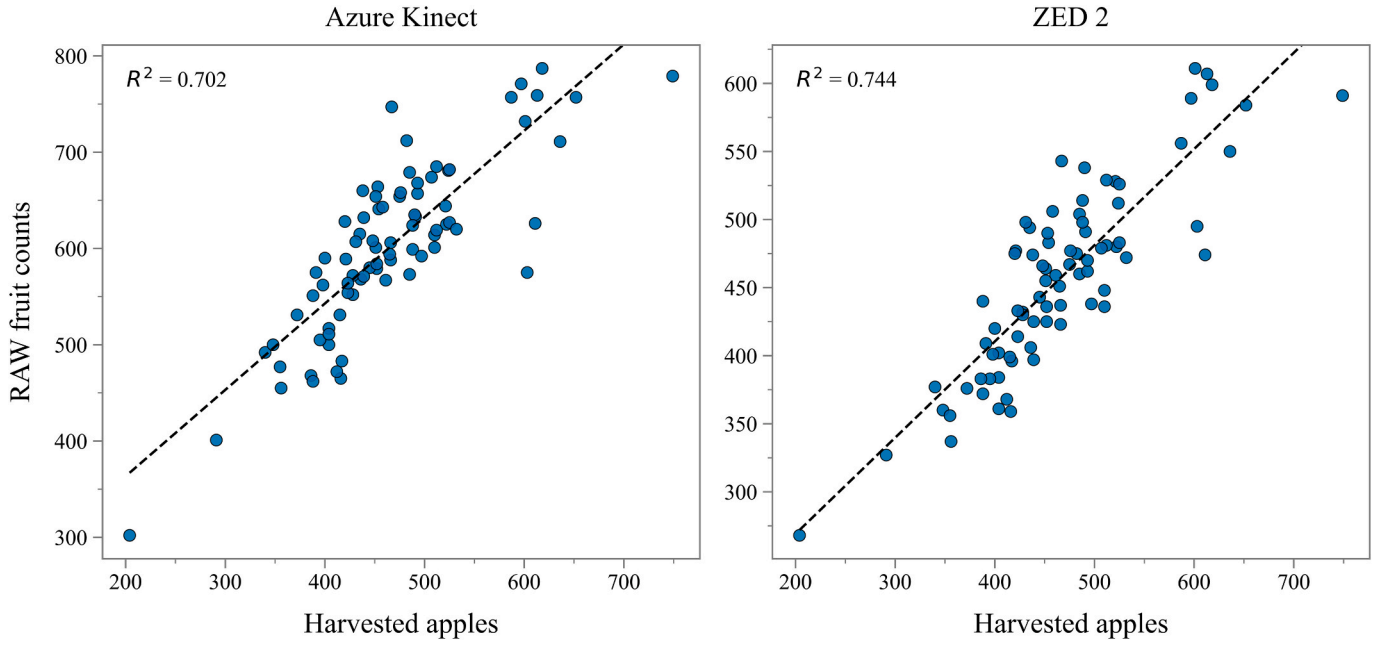

Fig. 7. Relationship between harvested apples (ground truth) and the raw fruit counts obtained using the load estimation pipeline; Azure Kinect (left) and ZED 2 (right).

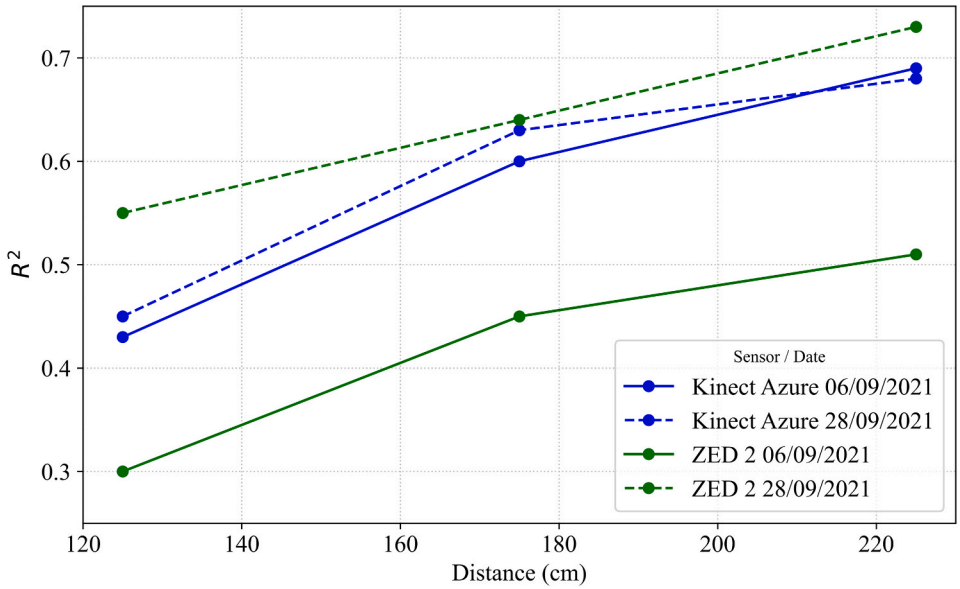

Fig. 8. Scanning performance by distance, comparison between ZED 2 and Kinect Azure camera.

(R^2) in the different configurations evaluated. The R^2 value is obtained by comparing the results of the system-estimated count with the ground truth. It can be seen that between the two acquisition dates, for the Azure Kinect, the value of R^2 does not vary too much for the different scanning distances. When the sensor is placed at 225 cm, the value of R^2 is 0.686 on 06/09 and 0.682 on 28/09. However, in the results obtained

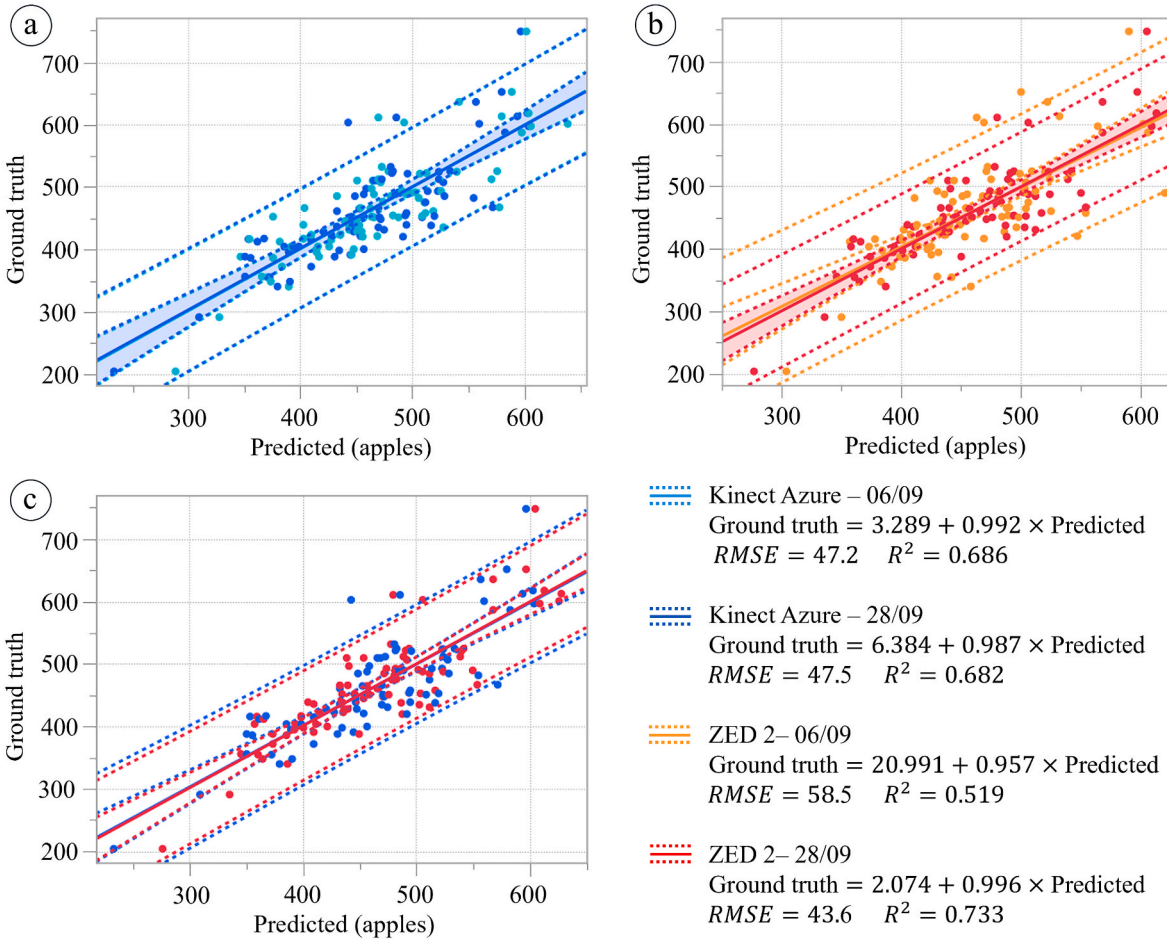

Fig. 9. Comparison of the dispersion graph and regression lines in three different situations, all them at 225 cm from the sensor with respect to the axis of the row of trees: a) Azure Kinect at two different dates; b) ZED 2 at two different dates; c) one week before the harvest, ZED 2 and Azure Kinect. Confidence and prediction bands are represented by dotted lines.

by the ZED 2 sensor there is a notable difference in the value of R^2 at the different acquisition dates. When the sensor is placed at 225 cm, R^2 values of 0.519 and 0.733, respectively, are obtained. This indicates a higher predictive capability when the data is captured on 28/09, less error as the harvest date is closer. It has been proved that the two sensors are affected by the variable 'distance', since they achieve a higher performance at a scanning distance of 225 cm compared to shorter distances. Therefore, it is established that for the following results only those obtained at a distance of 225 cm will be analysed.

Fig. 9 compares the regression lines between the actual number of fruits harvested and the number of fruits predicted with both sensors (Azure Kinect and ZED 2) and at different dates. Each of the graphs includes both confidence bands and prediction bands. The confidence band represents the uncertainty in the estimation of the mean value for different values of the predictor variable. On the other hand, the prediction band represents the uncertainty in the prediction of a new value for different values of the predictor variable.

Fig. 9a shows the results obtained with the Azure Kinect sensor at a distance of 225 cm, with the predictions for the scan conducted four weeks before harvest shown in light blue, and for the scan conducted one week before harvest shown in dark blue. The statistical analysis confirms that the two regression models are not significantly different, in fact, in the graph, the lines of the two models compared overlap. On the other hand, the uncertainty of both models is very similar, as the confidence bands (narrow) and the prediction bands (wide) are also practically overlapping, in addition to achieving a very close R^2 value between them. This indicates that the Azure Kinect sensor shows a certain temporal stability, making the predictive capability of the system practically unchanged whether the data is captured 4 weeks before harvest or 1 week before.

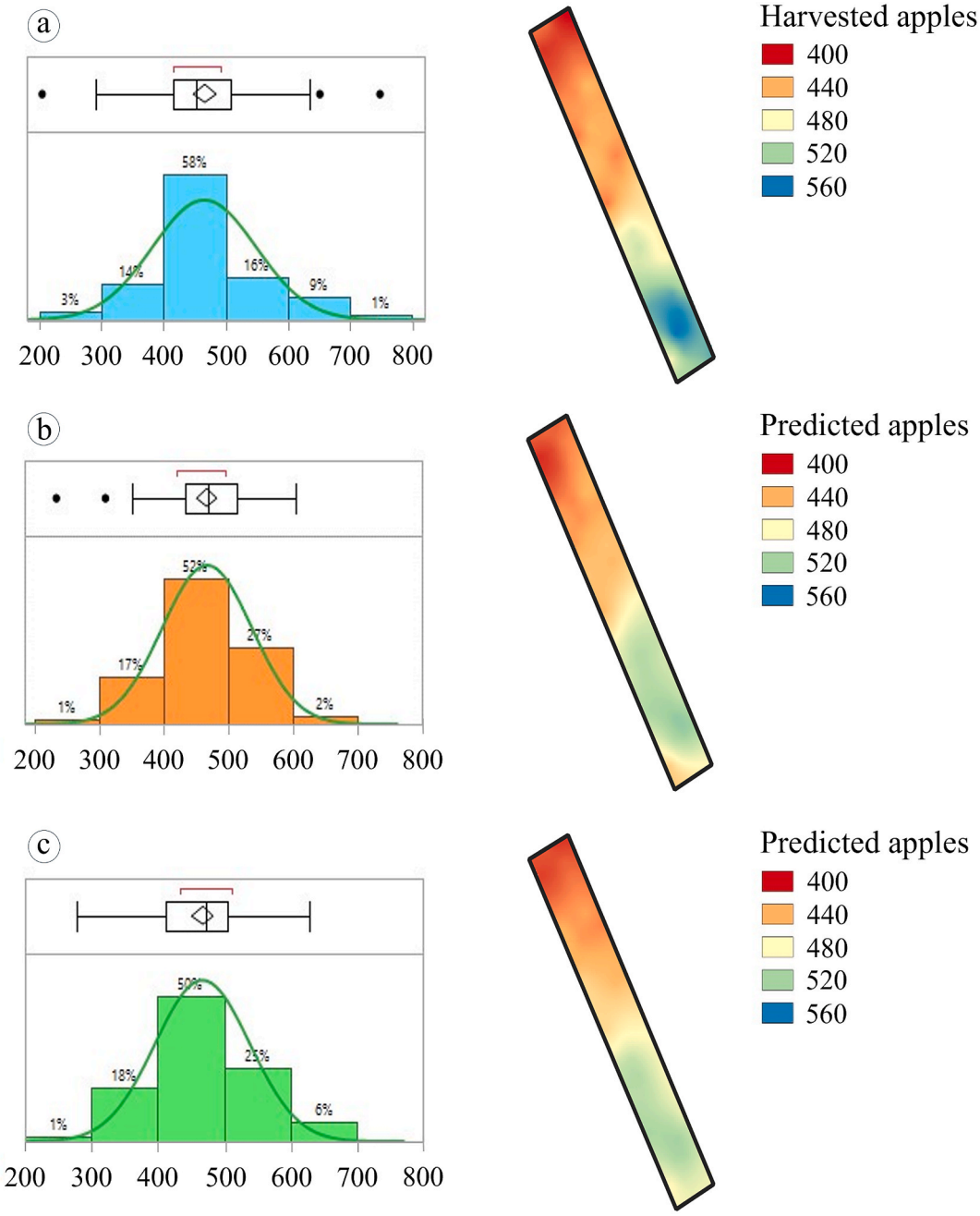

Fig. 10. Comparison of the distribution of predicted and actual values with their corresponding orchard block-level yield maps: a) ground truth; b) predictions with the Azure Kinect sensor; c) predictions with the ZED 2 sensor.

Fig. 9b presents the results from the ZED 2 sensor at a distance of 225 cm, where predictions from the scan taken four weeks before the harvest are displayed in orange, and predictions from the scan one week before the harvest are displayed in red. Although the regression models are not statistically different as the lines are practically overlapping, the uncertainty on the predictions is not the same. Both the confidence and prediction bands scanned one week before the harvest (in red) are narrower than those in orange, also achieving a better R^2. This indicates that with the ZED 2 sensor there is some temporal instability, with the predictions made with the data set acquired one week before the harvest being more reliable, which puts this sensor at a disadvantage compared to the Azure Kinect.

The third graph (Fig. 9c**)** compares Azure Kinect and ZED 2 with the data obtained at a distance of 225 cm one week before the harvest. In this case the two models are not statistically different, moreover, the uncertainty of their predictions is very similar, although the R^2 value is a little better with ZED 2, which explains why the red prediction bands are a little narrower.

Fig. 10 provides a comparison between the distribution of the harvested and predicted fruit counts and their corresponding orchard block-level yield maps.

The histogram in Fig. 10a shows the distribution of the harvested apples. A concentration of the apples is observed in the range 400 to 500, with a mean close to 450 apples. This suggests that most areas of the plantation have a similar density of apples.

Fig. 10b shows the distribution of the predicted data obtained with the Azure Kinect sensor at a distance of 225 cm, scanned one week before harvest. It presents a mean number of apples close to that of the ground truth, but shows a larger scatter, indicating that the model predicts values over a wider range. This is reflected in the outliers present in the boxplot. The corresponding orchard block-level yield map shows a similar spatial distribution to the ground truth, although with some discrepancies.

Fig. 10c illustrates the distribution of the predicted data obtained with the ZED 2 sensor at a distance of 225 cm, scanned one week before harvest. It has a slightly higher mean than the groundtruth, and the scatter is similar to that of the real data. As in Fig. 10b, the orchard block-level yield map shows a similar spatial distribution to the ground truth, although with some discrepancies. In particular, the highest-density regions are underrepresented, especially in Fig. 10c, where areas corresponding to the highest fruit load are less frequent than in the ground truth map. This suggests that the system captures the general spatial trend of fruit load, but tends to smooth local high-density peaks.

The results obtained by the two sensors are very similar, as both show to be consistent in the distribution of the load along the orchard. With the results analysed, it is observed that the ZED 2 sensor is less robust than the Azure Kinect, as it is affected by more variables, such as the scanning distance and the date of data acquisition. Given this condition, it is established that for the following results, only those captured with the Azure Kinect at 225 cm will be analysed.

3.3. Scanning 1 side vs scanning 2 sides (Kinect data – distance 225 cm)

To evaluate the influence of the scanning side (East, West or both) and data acquisition date on the system counting predictions, in Table 5, an Analysis of Variance (ANOVA) analysis has been performed.

For the MAE, MAPE, and R^2 metrics, there are no statistically significant differences for any of the factors evaluated (‘Date’ and ‘Side’), nor for their interaction, as all p-values are greater than 0.05. These results suggest that neither the timing of the data acquisition (four weeks before harvest or one week before) nor the side of the scan significantly influenced MAE, MAPE, or R2 under the evaluated conditions.

In contrast, for the RMSE metric, the 'Side' factor showed a significant effect, with a p-value of 0.024. This indicates that the effect of scanning side was mainly reflected in the magnitude of larger prediction errors, rather than in a consistent improvement across all evaluation metrics.

The boxplot in the Fig. 11 shows the distribution of RMSE values for the three scanning sides: Both, East, and West.

When evaluating the boxplot ‘A’, obtained from scanning both sides of the row, the distribution is relatively narrow, indicating a consistent performance with low variability in the RMSE values. The boxplot ‘A’ (All sides) shows a higher variability compared to the ‘E’ (East side) configuration, with a more dispersed range of RMSE values. The distribution for the ‘W’ (West side) shows the highest variability and has the most pronounced dispersion of the data.

The results indicate that scanning the row from both sides significantly reduced RMSE ($p = 0.0249$). However, since no significant differences were observed for MAE, MAPE, or R^2, this effect should be interpreted as a reduction in larger prediction errors or error variability,

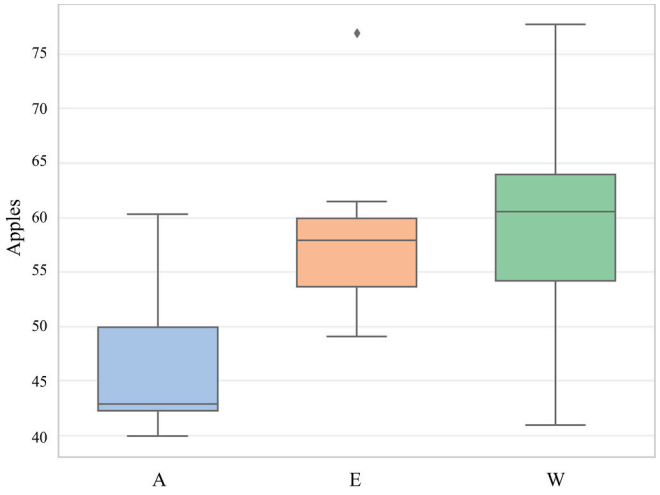

Fig. 11. RMSE distribution for each measurement side: Both (A), East (E) and West (W).

Table 5
Summary of ANOVA results for MAE, MAPE and R^2.

Metric	Source	Number of parameters	Degrees of freedom	Sum of squares	F ratio	Prob > F
MAE(apples)	Date	1	1	209.3	2.321	0.145
	Side	2	2	586.7	3.253	0.062
	Date*Side	2	2	96.3	0.534	0.595
MAPE(%)	Date	1	1	0.0014	2.425	0.136
	Side	2	2	0.0032	2.870	0.082
	Date*Side	2	2	0.0005	0.460	0.638
RMSE(apples)	Date	1	1	179.9	2.034	0.170
	Side	2	2	807.9	4.566	**0.024***
	Date*Side	2	2	108.8	0.615	0.551
R^2	Date	1	1	0.0049	0.121	0.731
	Side	2	2	0.0184	0.229	0.797
	Date*Side	2	2	0.0085	0.105	0.900

rather than as a comprehensive improvement across all statistical indicators. The adjusted RMSE (least squares mean) for the dual-side configuration was 46.73 apples, compared to 59.36 and 58.69 apples when scanning only from the west or east sides, respectively.

4. Discussion

4.1. Detection and tracking method selection

The developed counting system showed to be consistent in count and load distribution along the orchard. However, variability in the results was observed due to multiple factors, including tracker algorithm, sensor type, scanning date, distance, and the scanned side of the row.

The performance of three state-of-the-art tracking methods, SORT, DeepSORT, and ByteTrack, was compared for fruit counting in recorded videos. ByteTrack demonstrated higher scores in terms of MOTA, IDF1, and HOTA values, while maintaining a processing time comparable to SORT (Gené-Mola et al., 2023a; Matos et al., 2024). These results can be explained by ByteTrack's two-stage association strategy, which uses lower-confidence detections to recover unmatched trajectories and reduce missed detections and track fragmentation. This is particularly relevant in orchard images, where fruit detections can be affected by partial occlusions, viewpoint changes, and variable illumination.

In contrast, the appearance features employed by the DeepSORT algorithm did not improve fruit tracking (Zhang et al., 2024). This may be due to the inherent similarity in appearance among apples, which limits the effectiveness of feature-based Re-ID approaches for distinguishing individual fruits. In addition, DeepSORT required substantially higher processing time per image, averaging 128.0 ms compared with 15.3 ms for SORT and 15.4 ms for ByteTrack.

The detection and tracking components adopted in this study, YOLOv5 and ByteTrack, were state-of-the-art methods when the project began in 2022. Since then, newer architectures have reported improved results on general-purpose benchmarks, including recent iterations of the YOLO family such as YOLO26 (Ultralytics, 2026), and trackers that extend the tracking-by-detection paradigm with appearance-based re-identification present (Re-ID) and camera motion compensation (CMC), such as BoT-SORT (Aharon et al., 2022) and StrongSORT (Du et al., 2023). As a complementary check on whether these advances would benefit the task, a YOLO26 model was fine-tuned on the same annotated dataset used to train YOLOv5 and evaluated under the same conditions. No advantage was observed: performance was marginally lower than YOLOv5 (F1: 0.819 vs. 0.834; mAP@0.5: 0.865 vs. 0.871), indicating that gains reported for newer YOLO iterations on general-purpose benchmarks do not necessarily transfer to small, densely clustered and visually homogeneous fruit such as apples. A possible explanation is that, for this single-class task with a comparatively modest fine-tuning dataset, the remaining detection error is dominated by irreducible factors such as heavy occlusion and ambiguous fruit boundaries rather than by detector capacity, although a detailed analysis of this behaviour was beyond the scope of this study.

Regarding tracking, BoT-SORT and StrongSORT were additionally evaluated under the same conditions. BoT-SORT achieved the best tracking metrics, improving MOTA by approximately 4 %, IDF1 by 3 % and HOTA by 6 % with respect to ByteTrack. This gain is mainly attributable to its camera motion compensation, which corrects the Kalman filter prediction for the displacement of the vehicle-mounted sensor, but it came at approximately five times the processing time per image. StrongSORT, in contrast, did not outperform ByteTrack despite a substantially higher computational cost, suggesting that appearance-based Re-ID contributes little for objects as small and visually similar as apples, consistent with the DeepSORT results discussed above. Considering the current trend in precision agriculture towards real-time processing on edge devices, and that at the moderate, constant scanning speed used in this study trajectory fragmentation was not a practical concern, ByteTrack was considered to offer the best accuracy–efficiency trade-off and was retained for the counting pipeline. More broadly, the contribution of this work should be understood not as an algorithmic innovation, but as a systematic evaluation of an MOT-based fruit counting pipeline, validated against harvest-level ground truth, under a wide range of realistic acquisition conditions.

4.2. Effect of acquisition conditions

The fruit counting system was evaluated at different conditions. Our analysis showed that scanning distance plays a key role in accuracy, with both Azure Kinect and ZED 2 showing higher errors when scanning at closer distances. This effect is probably due to a reduced field of view (FoV), which limits the visibility of the upper and lower sections of the trees, resulting in undetected fruits and a higher number of tracklets in the tracking algorithm. In this context, the best results were obtained at 225 cm, as this distance offered an optimal trade-off between capturing the full tree canopy and maintaining sufficient resolution to clearly identify individual apples. At this range, the entire structure was visible within the field of view, reducing missed detections. However, this result is influenced by the tree architecture present in the studied orchard. In this case, the trees were trained using a fruiting wall system, with a maximum canopy height of 3.5 m, which favours a relatively continuous and planar canopy structure. For orchards with different training systems, the optimal sensor distance may differ, this should be considered when interpreting the results.

Evaluating the influence of scanning date, Azure Kinect maintained stable results across both different dates ($R^2 \approx 0.68$; MAPE $\approx$ 7.5–7.7 % at 225 cm), while ZED 2 has shown a significant improvement closer to harvest (R^2 increasing from 0.519 to 0.733; MAPE decreasing from 9.44 % to 6.91 %). This discordance seems to be related to changes in fruit colour, as apples turned from green to red closer to harvest, increasing colour contrast with the canopy. Although the overall counting results reflect the combined effects of detection, tracking, and canopy visibility, the temporal variation observed for the ZED 2 suggests that part of the error originates from detection sensitivity to visual conditions rather than from tracking instability or geometric factors alone. In addition, differences between sensors may also contribute to the observed variability. The Azure Kinect provides higher native image resolution, while the ZED 2 has a wider field of view and uses wide-angle optics, which may reduce the effective pixel density per fruit and introduce stronger geometric distortions toward the image borders. These factors can influence detection robustness under different acquisition conditions.

The effect of scanning side was also analysed through ANOVA, results showed that scanning from one or both sides of the row had a statistically significant influence on the RMSE results ($p = 0.024$), whereas MAE, MAPE and R^2 were not significantly affected. This pattern is consistent with the definition of RMSE, which disproportionately emphasizes large individual errors. In our dataset, the largest residuals were probably linked to trees affected by strong canopy occlusion or very high fruit load, situations in which one side tended to compensate for occlusions present in the other. By combining both viewpoints, these extreme deviations were notably reduced, while the central part of the error distribution remained largely unchanged. Consequently, RMSE detected a clear improvement when both sides were used, indicating that dual-side scanning is particularly effective in mitigating the most severe occlusion-related deviations. From a practical perspective, scanning only one side may be enough for obtaining a representative yield estimate, while scanning both sides mainly contributes to reducing the largest, occlusion-related prediction errors, rather than systematically improving all accuracy metrics. Depending on the intended use, growers may either scan a representative row from both sides or conduct full orchard scans accordingly. It should be noted that no cross-view identity matching or de-duplication between east and west scans was implemented, therefore, the improvement observed in dual-side configurations is attributed to complementary canopy visibility.

4.3. Robustness of the counting pipeline

Tracking errors such as missed detections, identity switches, or fragmented tracks may introduce local inconsistencies in the intermediate fruit trajectories used by the stretch assignment algorithm. However, the impact of these errors on the final yield estimation is partially mitigated by several aspects of the proposed pipeline. First, the stretch assignment relies on observations across multiple consecutive frames, and the final stretch label is determined through a majority-vote strategy, reducing the influence of occasional tracking inconsistencies. Second, fruit counts are aggregated at the stretch level (five-tree segments), which spatially averages local errors and limits the impact of individual tracking mistakes. Finally, the regression model used to estimate fruit load compensates for systematic observation biases caused by occlusions, duplicated observations from opposite sides of the row, and fragmented tracks. Nevertheless, since these error sources are compensated jointly, their individual contributions cannot be isolated from the corrected results, and a detailed analysis of how each of them (occlusion, duplicate observations, and tracking errors) propagates through the counting pipeline remains an interesting direction for future work.

Virtual counting line approaches have also been used in tracking-based fruit counting pipelines to reduce duplicate counts by counting fruits only when their trajectories cross a predefined line (Shen et al., 2023). However, in orchard environments, fruits may be partially occluded when crossing the virtual line, leading to missed counts. In contrast, the approach adopted in this study exploits the full camera field of view and the complete video sequence, providing multiple opportunities to detect and track each fruit across consecutive frames.

It should also be noted that the raw fruit counts obtained from the video sequences represent a partial observation of the fruit load rather than a direct measurement. Due to occlusions within the canopy, the limited field of view of the cameras, and duplicate observations of fruits visible from both sides of the row, systematic differences between the observed and actual number of fruits are expected. Consequently, the raw counts should be interpreted as a proxy of fruit load rather than an exact count. The regression model introduced in the methodology therefore acts as a bias-correction step that translates these video-based observations into reliable yield estimates at the stretch level.

Direct numerical comparison of counting accuracy across studies remains challenging, as datasets are typically acquired under different orchard structures and acquisition conditions. Moreover, in many previously reported fruit counting pipelines, the ground truth corresponds to fruits manually annotated in the images or videos used for evaluation. Very high correlations, with R^2 values frequently above 0.95 and counting accuracies around 90 % or higher, are achieved with this approach (Cao et al., 2025; Gao et al., 2022; Yang et al., 2024). While this methodology facilitates benchmarking of detection performance, it does not necessarily reflect the actual number of fruits in the orchard. In contrast, the present fruit estimation pipeline has been assessed against the actual number of fruits in the orchard, through in-field manual counting of all fruits after harvest. This approach is more representative from an end-user perspective, as it provides a more realistic assessment of the true counting capability of the developed pipeline. Therefore, the results of the present work cannot be directly compared with those from studies evaluated against image-based ground truth.

An important strength of the proposed system lies in its evaluation under realistic acquisition scenarios. The experiments considered different sensor types, scanning distances, acquisition dates, and scanning sides, allowing the robustness of the counting approach to be assessed under practical operational conditions rather than under a single fixed configuration. This experimental design provides a more representative understanding of the system performance in orchard conditions.

To assess the robustness of the proposed counting system, several methodological elements were incorporated. The regression models were evaluated using cross-validation across orchard rows, ensuring that performance metrics were computed on independent data. The regression analysis also included confidence and prediction intervals, which quantify model and prediction uncertainty. In addition, an ANOVA was conducted to examine whether differences in performance across acquisition conditions were statistically significant. Although full independent field repetitions were not possible, the use of two acquisition dates and two scanning sides provided repeated observations under diverse conditions, allowing us to examine the stability of the system across scenarios.

4.4. Limitations and future outlook

In the present study the stretch-level validation required the manual annotation of reference frames and the placement of physical markers to align video detections with harvested ground truth. Although this approach ensured precise evaluation at a 5-meter resolution, it is not intended for large-scale commercial deployment. In practical applications, fruit counting systems typically generate continuous yield estimates or aggregate counts at row or orchard scale, eliminating the need for predefined stretch segmentation or manual frame annotation. Future work could explore fully automated geo-referenced mapping strategies integrating GNSS and odometry for large-scale deployment.

At the same time, some limitations should be acknowledged. One limitation of this study is that all experiments were conducted within a single orchard with relatively homogeneous tree architecture, planting density, and management practices. Although the detection model was trained using annotated frames from a limited section of one row and evaluated across different rows of the orchard, demonstrating generalization within the same orchard, the variability between rows is likely smaller than the variability that may occur across different orchards. Therefore, further studies evaluating the proposed system under a wider range of orchard conditions, including different canopy structures, illumination environments, varieties, and fruit densities, would be valuable to fully assess the robustness and generalization capability of the detection model.

Finally, the increasing accessibility of these devices in recent years may soon enable many growers to acquire and install them on their tractors. This would allow data collection during each treatment application, from both sides of the crop row and at regular intervals. Such continuous monitoring would progressively enhance data accuracy and provide insights into crop development over time, equipping farmers with more detailed information to optimize management practices and anticipate potential issues.

5. Conclusions

This work presents the development of a computer vision system for fruit counting and load estimation, with a particular focus on evaluating its performance under varying scanning conditions. The system was tested in an experimental apple orchard to assess its behaviour under different acquisition configurations. Data was recorded using two RGB-D sensors (Azure Kinect DK and ZED 2) at three different distances (125 cm, 175 cm, and 225 cm). The orchard rows were scanned from both sides on two different dates (four weeks and one week prior to harvest). The developed system includes several stages: fruit detection, fruit tracking, assignment of each fruit to its corresponding stretch or region of the orchard, and the generation of an orchard block-level yield map. Within the evaluated experimental conditions, the Azure Kinect sensor showed more stable performance than the ZED 2, as it achieves more similar results on both dates. Furthermore, increasing the scanning distance up to 225 cm improved the balance between canopy visibility and fruit detection reliability in the studied fruiting wall orchard, leading to a MAPE of 6.91 % and an R^2 of 0.733 under that configuration. In addition, bilateral scanning improved fruit load estimation at the stretch level by integrating information from both sides of the canopy. However, this improvement was mainly reflected in RMSE, suggesting a

reduction in larger prediction errors or error variability, rather than a comprehensive improvement across all evaluation metrics. Moreover, since east and west side scans were processed independently, without instance-level cross-view matching or de-duplication, this improvement should be interpreted as an enhancement of stretch-level yield estimation rather than a direct increase in instance-level counting accuracy. Therefore, these findings should be interpreted in the context of the specific orchard architecture and acquisition setup considered in this study.

The results demonstrate the potential of the proposed system as a practical tool for improving fruit load estimation in apple orchards, it offers growers a more efficient and precise alternative to traditional manual methods. Generating detailed load maps provides growers objective and spatially localized information which is important for optimizing harvest logistics. The use of such systems can lead to increase the orchard profitability by improving the resource use efficiency compared to traditional estimation methods. Future work should validate the proposed system in different orchards, canopy architectures, and management conditions to assess its generalization beyond the same-orchard conditions considered here. Multi-view fusion strategies based on 3D fruit localization or cross-view matching should also be investigated to prevent double counting. Finally, integrating fruit size estimation would provide information on fruit size distribution within the orchard.

CRediT authorship contribution statement

Marc Felip-Pomés: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation. **Jordi Gené-Mola:** Writing – review & editing, Writing – original draft, Validation, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Jaume Arnó:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Jaume Lordan:** Writing – review & editing, Writing – original draft, Resources, Data curation. **Javier Ruiz-Hidalgo:** Writing – review & editing, Writing – original draft, Software. **Josep-Ramon Morros:** Writing – review & editing, Writing – original draft, Software. **Miguel Arasanz-Malo:** Formal analysis, Software, Validation, Writing – review & editing. **Francesc Net-Barnés:** Software, Formal analysis, Data curation. **Eduard Gregorio:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used ChatGPT-4 to improve readability and language. After using this tool, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The code and the dataset used to evaluate the method have been made publicly available at https://github.com/imatge-upc/VideoFruitCounting.

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